

Hysterical Response in Open Quantum Systems

Response theory, basis covariance, neutrality, instability, and infrared equivalence

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

We develop the open-system mathematics of the Hysterical Response: the residual response of an already-existing interaction under exponentially stable reduced dynamics. After setting the response functional and Newtonian channel geometry, we prove basis covariance relative to a physical classicalization map, so that “off-diagonal” language is representation-relative while the scalar response is invariant. We then give the constructive resolvent formulae, prove the Hysterical Neutrality Theorem and a gravitational survival corollary, package gravitational positivity at a stable supercritical saturated node, classify neutral/critical/unstable closed-loop regimes, and show that semiclassical and quantum-gravitational routes share the same weak-field infrared phenomenology when they retain the same potential.

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1 Introduction

Decoherence makes distinct quantum alternatives (informally, Everett branches) behave, for local observations, as if they were separate classical possibilities. Coherent interference among those alternatives is erased from the reduced description; a residual, history-dependent contribution to the response of ordinary forces under open-system dynamics can remain. Hysterical Response is that residual contribution: a dynamical modification of an already-existing interaction. The response exhibits *hysteresis*—dependence on its history. The correct adjective would be *hysteretic*, but we have chosen *Hysterical*, since life is short, and whimsy is important. This paper supplies the mathematical core of that idea.

Statements are labelled as follows.

Theorem	Consequence of the mathematical assumptions stated here.
Modeling assumption	Closure connecting the abstract response to a physical channel or order parameter.
Physical input	Property of the underlying interaction, not derived here.
Conjecture	Proposed extension requiring further microscopic derivation.

No Everett branch count or branch metric is assumed. The reduced dynamics are those of an open quantum system in the sense of quantum dynamical semigroups [3, 4, 5]; classicalization structure is motivated by environment-induced superselection [6, 7].

2 Setup: states, dephasing, and Newtonian geometry

2.1 State space, localization, and force observables

Assumption 2.1 (State space). Let $\mathcal{H}$ be a separable complex Hilbert space. States are positive trace-class operators $\rho \in \mathcal{T}_1(\mathcal{H})$ with $\text{Tr } \rho = 1$.

Definition 2.2 (Localization dephasing). A physically selected localization/configuration structure is represented by mutually orthogonal projections $\{E_a\}$ with $\sum_a E_a = I$ strongly, or by a spectral measure together with a finite-resolution measurable partition. Define

$$\Delta_{\text{loc}}(X) = \sum_a E_a X E_a, \quad (1)$$

with trace-norm convergence on the trace class. A second dephasing map Δ_{coh} may be specified when the model asks for coherence relative to a conjugate, mixed, or otherwise physically chosen diagnostic decomposition; it need not coincide with Δ_{loc} .

Definition 2.3 (Force expectation and dephasing-relative coherence). For a bounded vector force $F \in \mathcal{B}(\mathcal{H})^3$ define

$$F(\rho) = \text{Tr}(F\rho), \quad F_{\text{coh}}^{(\Delta)}(\rho) = \text{Tr}[F(I - \Delta)\rho]. \quad (2)$$

Thus $F_{\text{coh}}^{(\Delta)}(\rho) = F(\rho) - F(\Delta\rho)$. For unbounded force operators, subsequent statements are understood on an admissible domain on which the indicated force pairings and resolvent integrals exist. In a finite-dimensional matrix representation with ordered basis $\{|i\rangle\}$, the entries are $F_{ij} = \langle i|F|j\rangle$: the *diagonal* pieces are the $i = j$ entries and the *off-diagonal* pieces are the $i \neq j$ entries. Relative to a decoherence-selected pointer basis, those off-diagonal blocks are the natural home for coherences between approximately classical alternatives, and “off-diagonal force” refers to that matrix structure (or its continuum analogue).

Proposition 2.4 (Dephasing no-go identity). *Let $\Delta^\dagger$ be the adjoint of Δ on observables. If $\Delta^\dagger(F) = F$, then $F_{\text{coh}}^{(\Delta)}(\rho) = 0$ for every admissible ρ .*

Proof. $\text{Tr}(F\Delta\rho) = \text{Tr}(\Delta^\dagger F\rho) = \text{Tr}(F\rho)$. □

Remark 2.5 (Why two structures may be needed). For a Newtonian force diagonal in the same localization algebra preserved by Δ_{loc} , the coherence contribution relative to Δ_{loc} vanishes identically. A nonzero “off-diagonal force” then refers either to a distinct diagnostic dephasing Δ_{coh} , or to a transformed representation in which the force is not fixed by that dephasing. The total source-induced force response defined below is independent of any separate coherence decomposition.

Lemma 2.6 (Ensemble linearity). *For any admissible ensemble,*

$$\mathbb{E} F(\rho_\omega) = F(\mathbb{E}\rho_\omega), \quad \mathbb{E} F_{\text{coh}}^{(\Delta)}(\rho_\omega) = F_{\text{coh}}^{(\Delta)}(\mathbb{E}\rho_\omega). \quad (3)$$

2.2 Newtonian geometry from particle number density

Definition 2.7 (Number density). $K(x) \geq 0$ denotes particles per unit volume. Hence

$$N = \int_{\mathbb{R}^3} K(x) \, d^3x. \quad (4)$$

For identical constituent mass m , the mass density is $\rho_m = mK$. For an observation point x ,

$$\Phi(x) = -Gm \int \frac{K(y)}{|x-y|} \, d^3y, \quad g(x) = -Gm \int K(y) \frac{x-y}{|x-y|^3} \, d^3y. \quad (5)$$

A test particle of mass m_t experiences $f = m_t g$.

Assumption 2.8 (Force-integrable density). We assume $K \in L^1(\mathbb{R}^3)$ and, at each observation point used in the model,

$$\int_{|x-y|<1} \frac{K(y)}{|x-y|^2} \, d^3y < \infty. \quad (6)$$

Locally bounded K satisfies this; singular sheets or point components require a stated regularization.

2.2.1 Spherical number density

For $K(x) = K(r)$,

$$N = 4\pi \int_0^\infty K(r) r^2 \, dr, \quad M(< r) = 4\pi m \int_0^r K(s) s^2 \, ds, \quad (7)$$

and the shell theorem gives

$$g(r) = -\frac{4\pi Gm}{r^2} \left(\int_0^r K(s) s^2 \, ds \right) \hat{r}. \quad (8)$$

2.2.2 Axisymmetric cylindrical number density

For $K = K(r, h)$,

$$N = 2\pi \int_0^\infty r \, dr \int_{-\infty}^\infty K(r, h) \, dh. \quad (9)$$

At $(R, 0, Z)$ the exact components are

$$g_R(R, Z) = -Gm \int_0^\infty dr \int_{-\infty}^\infty dh \, r K(r, h) \int_0^{2\pi} \frac{R - r \cos \phi}{[R^2 + r^2 - 2Rr \cos \phi + (Z - h)^2]^{3/2}} d\phi, \quad (10)$$

$$g_Z(R, Z) = -Gm \int_0^\infty dr \int_{-\infty}^\infty dh \, r K(r, h) \int_0^{2\pi} \frac{Z - h}{[R^2 + r^2 - 2Rr \cos \phi + (Z - h)^2]^{3/2}} d\phi. \quad (11)$$

Equivalently,

$$\Phi(R, Z) = -4Gm \int_0^\infty dr \int_{-\infty}^\infty dh \frac{r K(r, h)}{\sqrt{(R + r)^2 + (Z - h)^2}} \mathcal{K}(k), \quad (12)$$

where $\mathcal{K}(k)$ is the complete elliptic integral of the first kind and

$$k^2 = \frac{4Rr}{(R + r)^2 + (Z - h)^2}. \quad (13)$$

Then $g = -\nabla \Phi$. If $K(r, h) = K(r, -h)$, then $g_Z(R, 0) = 0$.

2.3 Hamiltonian structure and gravitational channel data

Let $H = T + V$ and $F = -\nabla_q V$ for a chosen test/configuration coordinate q . For two configurations separated symmetrically by ℓ , define

$$\Delta = U(q - \ell/2) - U(q + \ell/2), \quad \delta f = f(q - \ell/2) - f(q + \ell/2). \quad (14)$$

If the required derivatives exist and are continuous near q ,

$$\Delta = \ell \cdot f(q) + O(|\ell|^3), \quad \delta f = -(\ell \cdot \nabla) f(q) + O(|\ell|^3). \quad (15)$$

Thus the local force contrast is controlled by a tidal gradient rather than automatically by the absolute field. (Here $f = -\nabla V$ when U denotes the same potential as V .)

2.4 Response functional

For an interaction force observable F , open-system generator $\mathcal{L}$, and admissible source $\mathcal{S}$ on a stable response subspace, the stationary Hysterical response is

$$\delta \rho_H = -\mathcal{L}^{-1} \mathcal{S}, \quad F_H = -\text{Tr}(F \mathcal{L}^{-1} \mathcal{S}). \quad (16)$$

No new force carrier is introduced: F_H is a component of the response of F itself. The constructive theory of Section 4 identifies $-\mathcal{L}^{-1}$ with the Laplace integral of the semigroup under exponential stability.

3 Basis covariance

The central distinction is

$$\boxed{\text{arbitrary coordinate basis} \neq \text{physical decoherence-selected classical structure.}} \quad (17)$$

No exact preferred orthonormal branch basis is required. In realistic systems the classical structure may be approximate, redundant, block-valued, or described by a pointer algebra rather than by exact rank-one projectors. For the theorem we use a linear idempotent classicalization map, which includes ordinary projective dephasing as the simplest case.

3.1 The physical classicalization map

Let $\mathcal{H}$ be a Hilbert space and let $\mathcal{T}_1(\mathcal{H})$ denote the trace-class operators. Let

$$\Delta_{\text{cl}} : \mathcal{T}_1(\mathcal{H}) \rightarrow \mathcal{T}_1(\mathcal{H}) \quad (18)$$

be a bounded linear, trace-preserving, idempotent map,

$$\Delta_{\text{cl}}^2 = \Delta_{\text{cl}}, \quad (19)$$

representing the approximately classical sector selected by environmental dynamics.

For a simple pointer decomposition with projectors P_a ,

$$\Delta_{\text{cl}}(X) = \sum_a P_a X P_a. \quad (20)$$

The corresponding off-classical projector on operator space is

$$Q_{\text{cl}} := I - \Delta_{\text{cl}}. \quad (21)$$

Definition 3.1 (Off-classical component). For any trace-class operator X , define

$$X_{\text{off}} := Q_{\text{cl}} X = (I - \Delta_{\text{cl}}) X. \quad (22)$$

When (20) applies, this coincides with the matrix off-diagonal part in the pointer representation. Equation (22), rather than matrix off-diagonality in an arbitrary coordinate basis, is the invariant definition used here.

Remark 3.2 (Everett terminology). The phrase “off the diagonal relative to classical-like Everett timelines” means precisely “outside the image of Δ_{cl} ,” with the latter supplied by decoherence. The formal results require neither a branch count nor a metric on branches, and they remain meaningful without Everett language.

3.2 Passive unitary changes of representation

For a unitary U define

$$\text{Ad}_U(X) := U X U^\dagger. \quad (23)$$

A passive change of representation carries every physical object into its new matrix representation. Thus

$$\rho' = \text{Ad}_U(\rho), \quad (24)$$

$$F' = \text{Ad}_U(F), \quad (25)$$

$$\mathcal{S}' = \text{Ad}_U(\mathcal{S}), \quad (26)$$

and superoperators transform by conjugation,

$$\Delta'_{\text{cl}} = \text{Ad}_U \circ \Delta_{\text{cl}} \circ \text{Ad}_{U^\dagger}, \quad (27)$$

$$\mathcal{L}' = \text{Ad}_U \circ \mathcal{L} \circ \text{Ad}_{U^\dagger}. \quad (28)$$

Lemma 3.3 (Covariance of the off-classical sector). *Let $Q_{\text{cl}} = I - \Delta_{\text{cl}}$ and $Q'_{\text{cl}} = I - \Delta'_{\text{cl}}$. Then for every trace-class X ,*

$$\boxed{Q'_{\text{cl}} \text{Ad}_U(X) = \text{Ad}_U(Q_{\text{cl}}X)}. \quad (29)$$

Proof. Using (27),

$$Q'_{\text{cl}} \text{Ad}_U(X) = \text{Ad}_U(X) - \Delta'_{\text{cl}} \text{Ad}_U(X) \quad (30)$$

$$= \text{Ad}_U(X) - \text{Ad}_U \Delta_{\text{cl}}(X) \quad (31)$$

$$= \text{Ad}_U[(I - \Delta_{\text{cl}})X]. \quad (32)$$

□

Thus an operator may cease to look matrix off-diagonal after a basis change, but it remains the same off-classical physical operator relative to the transformed classicalization map.

3.3 Basis covariance theorem for Hysterical Response

On the stable response subspace,

$$\delta\rho_H = -\mathcal{L}^{-1}\mathcal{S}_H, \quad F_H = -\text{Tr}(F\mathcal{L}^{-1}\mathcal{S}_H). \quad (33)$$

Here F is an existing interaction observable. The source $\mathcal{S}_H$ is *system-dependent*: its construction, support, and normalization depend on the interaction, environment, coarse-graining, and state under study. It may be fed by the off-classical sector, replenishment, correlations, or other open-system structure; no new fundamental force observable is introduced. The present paper develops the response calculus for an admissible $\mathcal{S}_H$; Sections 4.1–4.3 derive the driven equation and an off-classical source for unentangled incoming particles in a large- n Everett toy, while continuum normalizations remain model-dependent.

Assumption 3.4 (Covariant source construction). *If the entire physical description is passively transformed by U , the response source transforms as*

$$\mathcal{S}'_H = \text{Ad}_U(\mathcal{S}_H). \quad (34)$$

In particular, any source constructed covariantly from $Q_{\text{cl}}\rho$, the interaction, and environmental data satisfies this requirement.

Lemma 3.5 (Covariance of the inverse generator). *If $\mathcal{L}$ is invertible on the retained response subspace and $\mathcal{L}'$ is given by (28), then*

$$(\mathcal{L}')^{-1} = \text{Ad}_U \circ \mathcal{L}^{-1} \circ \text{Ad}_{U^\dagger}. \quad (35)$$

Proof. Composition of the right-hand side with $\mathcal{L}'$ gives the identity on the transformed response subspace, since $\text{Ad}_{U^\dagger} \circ \text{Ad}_U = I$. □

Theorem 3.6 (Basis Covariance Theorem for Hysterical Response). *Let U be unitary and transform $F, \mathcal{L}, \mathcal{S}_H$, and Δ_{cl} passively according to (26)–(34). Then*

$$\boxed{\delta\rho'_H = \text{Ad}_U(\delta\rho_H)} \quad (36)$$

and the predicted leftover force is invariant (componentwise if F is vector-valued),

$$\boxed{F'_H = F_H}. \quad (37)$$

Proof. By (35) and (34),

$$\delta\rho'_H = -(\mathcal{L}')^{-1}\mathcal{S}'_H \quad (38)$$

$$= -\text{Ad}_U \mathcal{L}^{-1} \text{Ad}_{U^\dagger} \text{Ad}_U(\mathcal{S}_H) \quad (39)$$

$$= \text{Ad}_U(-\mathcal{L}^{-1}\mathcal{S}_H). \quad (40)$$

For the observable response,

$$F'_H = -\text{Tr}[UFU^\dagger U(\mathcal{L}^{-1}\mathcal{S}_H)U^\dagger] \quad (41)$$

$$= -\text{Tr}[UF(\mathcal{L}^{-1}\mathcal{S}_H)U^\dagger] \quad (42)$$

$$= -\text{Tr}[F\mathcal{L}^{-1}\mathcal{S}_H], \quad (43)$$

using unitary invariance/cyclicity of the trace. $\square$

Corollary 3.7 (No arbitrary basis ambiguity). *A passive basis change cannot create, destroy, or alter a Hysterical response. It can only change how the same physical response is represented.*

3.4 What is basis-relative and what is physical

The theorem does *not* say that the words “diagonal” and “off-diagonal” are representation invariant. They are not. It says that the off-classical decomposition is covariant once the physical map Δ_{cl} is transformed together with the state.

Proposition 3.8 (Pointer-structure distinction). *Replacing Δ_{cl} by a different map $\tilde{\Delta}_{\text{cl}}$ while keeping the physical state and interaction fixed is not a passive basis change unless*

$$\tilde{\Delta}_{\text{cl}} = \text{Ad}_U \circ \Delta_{\text{cl}} \circ \text{Ad}_{U^\dagger} \quad (44)$$

for the same passive representation transformation applied to all physical objects. In general, changing Δ_{cl} changes the physical coarse-graining model.

This is the precise sense in which the classical-like Everett decomposition is physical but not a mathematically preferred coordinate basis. The environment selects the relevant approximately classical algebra; a theorist is free to represent that algebra in any basis.

3.5 Active versus passive unitaries

A passive unitary changes coordinates. An active unitary changes the physical state while leaving the measuring apparatus, force observable, and environmental structure fixed.

Definition 3.9 (Active transformation). For fixed $F, \mathcal{L}, \mathcal{S}_H$ -construction, and Δ_{cl} , an active transformation sends

$$\rho \mapsto U\rho U^\dagger \quad (45)$$

without simultaneously conjugating the rest of the physical description.

There is no invariance theorem for an active transformation. Indeed, physically moving a source from a near to a far configuration, or changing a correlation, should generally change the force response.

Operation		What changes	Physical prediction
Passive change	basis	all state/observable/dynamical objects covariantly	unchanged
Active unitary		state/configuration, with apparatus/dynamics fixed	may change
Pointer-map placement	re-	environmental classicalization structure	may change

3.6 Application: three-particle correlation diagnostics

Let A be a probe and B, C near/far sources. Compare

$$|\Psi_+\rangle = \frac{1}{\sqrt{2}}(|B_N C_N\rangle + |B_F C_F\rangle), \quad |\Psi_-\rangle = \frac{1}{\sqrt{2}}(|B_N C_F\rangle + |B_F C_N\rangle). \quad (46)$$

Both states have identical one-body marginals $P(B_N) = P(B_F) = P(C_N) = P(C_F) = \frac{1}{2}$, yet opposite near/far correlation. With equal source masses and a symmetric geometry, writing F_N and F_F for the force contribution of one near or far source at A , the branch force contrasts are

$$\Delta F_+ = 2(F_N - F_F), \quad \Delta F_- = 0. \quad (47)$$

Thus a history-dependent response can satisfy $F_H^{(A)}[\Psi_+] \neq F_H^{(A)}[\Psi_-]$ even though the separate statistics of B and C are unchanged. With $Z = |N\rangle\langle N| - |F\rangle\langle F|$ one has $\langle Z_B \rangle = \langle Z_C \rangle = 0$ while $\langle Z_B Z_C \rangle_\pm = \pm 1$; a correlation-sensitive probe observable therefore sees a nonzero discriminator $2r_{BC}$ between the two states. Under a simultaneous passive unitary change of representation that discriminator is unchanged (in a Hadamard frame it may appear as a relative phase). This is why Theorem 3.6 is required: “off-diagonal” language is relative to a physically selected classicalization map, not to an arbitrary coordinate basis.

3.7 Why “off-diagonal” remains useful

The preceding theorem does not require abandoning the original intuition. In the physically selected near/far pointer representation, the object

$$Q_{\text{cl}}\rho = (I - \Delta_{\text{cl}})\rho \quad (48)$$

is literally represented by off-diagonal blocks between approximately classical configurations. Thus the phrase “off-diagonal response” is legitimate shorthand when the pointer representation has already been specified.

The mature statement is therefore

Hysterical Response is the dynamical response of an existing interaction under open-system drive. When that drive is fed by structure outside the decoherence-selected approximately classical sector, the leftover is history-dependent in the sense of the present framework. Matrix diagonal vs. off-diagonal language is representation-dependent; the relation to the physical classicalization map is not.

(49)

This also clarifies the static dephasing no-go. The instantaneous quantity

$$\text{Tr}[F(I - \Delta_{\text{cl}})\rho] \quad (50)$$

may vanish when $\Delta_{\text{cl}}^\dagger(F) = F$. Hysterical Response is not identified with that instantaneous matrix element. Rather, a system-dependent source $\mathcal{S}_H$ —which *may* carry off-classical, correlational, or replenishment structure—produces

$$F_H = -\text{Tr}(F\mathcal{L}^{-1}\mathcal{S}_H), \quad (51)$$

which need not vanish even when the static coherence pairing does.

3.8 Open physical questions

The theorem resolves the representation problem but deliberately leaves the following system-dependent questions open:

1. the microscopic derivation of Δ_{cl} or the pointer algebra for a given environment;
2. how accurately the classicalization map is idempotent when decoherence is only approximate;
3. the concrete construction and normalization of the system-dependent source $\mathcal{S}_H$ in particular interactions and states (Section 4.1–4.3 give the unentangled-incoming / large- n Everett class; continuum amplitudes remain open);
4. the magnitude and spectrum of the associated Hysterical susceptibility;
5. whether the response is neutral, critical, or unstable under the instability criterion of Section 6;
6. whether a surviving response remains macroscopic after coarse graining.

None of these is an arbitrary-basis ambiguity.

4 Constructive response theory

4.1 Unentangled incoming particles and the driven response equation

The driven linear response equation used throughout this paper is not an independent postulate. We derive it for a generic open quantum system that decoheres while incorporating incoming degrees of freedom that arrive unentangled with the system.

Assumption 4.1 (Decoherence toward a classical reference). Let $\mathcal{L}_D$ generate a quantum dynamical semigroup on the trace class, and let ρ_0 be a classical reference state for a linear idempotent classicalization map Δ_{cl} :

$$\Delta_{\text{cl}}\rho_0 = \rho_0, \quad \mathcal{L}_D\rho_0 = 0. \quad (52)$$

Interpret $\mathcal{L}_D$ as the reduced generator of outgoing / monitored exchange (decoherence and relaxation toward the pointer sector).

Assumption 4.2 (Unentangled incorporation). An incoming degree of freedom with state σ_{in} is incorporated unentangled from the system,

$$\rho \longmapsto \rho \otimes \sigma_{\text{in}}, \quad (53)$$

then coupled by a joint unitary (or CPTP) interaction and reduced by tracing out the discarded outgoing registers. The induced system map

$$\mathcal{E}(\rho) = \text{Tr}_{\text{in}}[U(\rho \otimes \sigma_{\text{in}})U^\dagger] \quad (54)$$

is linear and completely positive for fixed σ_{in} . (As elsewhere in the paper, at incorporation we take the product state $\rho \otimes \sigma_{\text{in}}$; subsequent dynamics may entangle before the partial trace.)

Assumption 4.3 (Poisson arrivals). Between arrivals the reduced state evolves under $\mathcal{L}_D$. Arrivals occur at mean rate $\nu \geq 0$, yielding the piecewise-deterministic / jump-diffusion generator

$$\dot{\rho} = \mathcal{L}_D \rho + \nu(\mathcal{E}(\rho) - \rho) \quad (55)$$

on the system state.

Theorem 4.4 (Driven deviation equation). *Under Assumptions 4.1–4.3, write $\rho = \rho_0 + \delta\rho$. Then*

$$\dot{\delta\rho} = \mathcal{L} \delta\rho + \mathcal{S}, \quad (56)$$

where

$$\mathcal{L} = \mathcal{L}_D + \nu(\mathcal{E} - I), \quad \mathcal{S} = \nu(\mathcal{E}(\rho_0) - \rho_0). \quad (57)$$

Proof. Substitute $\rho = \rho_0 + \delta\rho$ into (55) and use linearity of $\mathcal{L}_D$ and $\mathcal{E}$ together with $\mathcal{L}_D \rho_0 = 0$:

$$\begin{aligned} \dot{\delta\rho} &= \mathcal{L}_D(\rho_0 + \delta\rho) + \nu(\mathcal{E}(\rho_0 + \delta\rho) - (\rho_0 + \delta\rho)) \\ &= \mathcal{L}_D \delta\rho + \nu(\mathcal{E}(\delta\rho) - \delta\rho) + \nu(\mathcal{E}(\rho_0) - \rho_0) \\ &= \mathcal{L} \delta\rho + \mathcal{S}. \end{aligned}$$

□

Thus the additive source $\mathcal{S}$ is precisely the mean kick that one unentangled arrival imparts to the classical reference ρ_0 , while $\mathcal{L}$ collects decoherence together with the linearized arrival map. If $\mathcal{E}(\rho_0) = \rho_0$, then $\mathcal{S} = 0$ and (56) reduces to homogeneous relaxation of deviations.

Remark 4.5 (Literature status of the driven equation). Affine / inhomogeneous open-system equations of the schematic form $\dot{\rho} = \mathcal{L}\rho + \mathcal{S}$ (equivalently $\dot{\delta\rho} = \mathcal{L}\delta\rho + \mathcal{S}$ after centering on a reference) are standard in the driven–dissipative and linear-response literature [5, 11]. What is derived here is not the existence of that form, but its identification with unentangled arrivals plus decoherence, and (below) the large- n Everett reason for an off-classical $\mathcal{S}$.

4.2 Incoming / outgoing fluxes and the $J_{\text{in}}/J_{\text{out}}$ scaling

Write J_{in} for a characteristic incoming particle rate (so $J_{\text{in}} \propto \nu$ in Assumption 4.3) and J_{out} for a characteristic outgoing / monitored exchange rate that sets the decoherence scale in $\mathcal{L}_D$.

Proposition 4.6 (Source scales with incoming flux). *Under Theorem 4.4,*

$$\mathcal{S} = J_{\text{in}} \mathcal{S}_1, \quad \mathcal{S}_1 := \frac{\nu}{J_{\text{in}}}(\mathcal{E}(\rho_0) - \rho_0), \quad (58)$$

whenever $J_{\text{in}} \propto \nu > 0$. In particular $\|\mathcal{S}\|_1 \propto J_{\text{in}}$ at fixed per-particle kick $\mathcal{E}(\rho_0) - \rho_0$. Thus $\mathcal{S}$ itself tracks incoming flux, not the ratio $J_{\text{in}}/J_{\text{out}}$.

Assumption 4.7 (Outgoing flux sets decoherence scale). The stable response rate of $\mathcal{L}_D$ (and hence of $\mathcal{L}$ when arrivals are not faster than decoherence) is of order the outgoing / monitored particle scale:

$$\gamma \sim J_{\text{out}}. \quad (59)$$

Remark 4.8 (Literature status of $\gamma \sim J_{\text{out}}$). This scaling is standard in environmental decoherence. In collisional decoherence, the localization rate is proportional to the environmental particle flux times an effective cross section [8, 9, 10] (surveyed in [7]). In emission / loss Lindblad models the same jump parameter γ that damps coherences sets the mean outgoing particle flux [5, 11]. Assumption 4.7 records that shared scale; it does not assert the numerical identity “decoherence rate = particle current.”

Corollary 4.9 (Steady leftover scales as $J_{\text{in}}/J_{\text{out}}$). *Under Theorem 4.4, Proposition 4.6, Assumption 4.7, and the exponential stability bound of Lemma 5.2,*

$$\|\delta\rho_{\text{ss}}\|_1 = \|\mathcal{L}^{-1}\mathcal{S}\|_1 \lesssim \frac{M}{\gamma} \|\mathcal{S}\|_1 \sim \frac{J_{\text{in}}}{J_{\text{out}}}, \quad (60)$$

and likewise $|F_H| \lesssim (M/\gamma) \|F\|_\infty \|\mathcal{S}\|_1$ scales as $J_{\text{in}}/J_{\text{out}}$ at fixed per-particle kick and force norm.

Remark 4.10 (Lean status). Lean packages the resolvent scaling as `cor_jin_over_jout`, with $\gamma \sim J_{\text{out}}$ taken as the tagged physics hypothesis `physics_jout_sets_decoherence_scale`.

4.3 Finite-dimensional Everett toy: large- n off-diagonal feed

We now specialize to a finite-dimensional Everett interpretational toy that explains why $\mathcal{S}$ is a *coherence* / off-classical drive rather than a reshuffling of branch populations.

Let $\dim \mathcal{H} = n < \infty$, and let $\{|i\rangle\}_{i=1}^n$ be a decoherence-selected Everett pointer basis, with

$$\Delta_{\text{cl}}\rho = \sum_{i=1}^n |i\rangle \langle i| \rho |i\rangle \langle i|. \quad (61)$$

Diagonal entries are the populations of the n alternatives; off-diagonal entries are coherences between them. Outgoing coupling to environment registers implements $\mathcal{L}_D$ and drives the reduced state toward the diagonal sector. An incoming particle is a finite ancilla prepared in σ_{in} , incorporated as in Assumption 4.2.

Assumption 4.11 (Large- n diagonal protection). In the Everett pointer basis, for large n , an unentangled incoming incorporation does not change the diagonal part of the system state at leading order: the information it contributes is not a $O(1)$ redistribution of branch populations. Equivalently, on the classical reference,

$$\Delta_{\text{cl}}(\mathcal{E}(\rho_0)) = \Delta_{\text{cl}}\rho_0 = \rho_0. \quad (62)$$

Interpretation. With many alternatives, a single incoming degree of freedom cannot preferentially repopulate one Everett branch over another at macroscopic order; any population shift is diluted across the pointer basis. The new quantum information therefore appears in the off-diagonal (coherence / correlation) sector.

Toy picture (outgoing bits versus incoming entanglement). In a finite-dimensional pointer / Hadamard toy, each bit of which-path information carried away by an outgoing environmental degree of freedom splits alternatives: the reduced dynamics is driven toward the classical diagonal, and interference between Everett branches is suppressed. An unentangled incoming degree of freedom can

restore as much quantum information in magnitude, but not by selecting one branch over another. Schematically, a shared ancilla state $|\phi\rangle$ attached symmetrically,

$$|\phi\rangle \otimes (|\text{branch}_1\rangle + |\text{branch}_2\rangle) \longrightarrow |\phi\rangle|\text{branch}_1\rangle + |\phi\rangle|\text{branch}_2\rangle, \quad (63)$$

puts the new information into *entanglement / coherence between* the branches: both alternatives “see” the same ϕ , so neither pointer population is preferentially fed. That is precisely an off-classical drive of the kind Assumption 4.11 isolates, and it is the toy reason the residual source scales with incoming flux J_{in} while decoherence / outgoing monitoring scales with J_{out} . The picture is interpretational bookkeeping for the large- n toy—not a continuum QFT derivation, and not a claim that every outgoing interaction is a literal one-bit Hadamard split.

Theorem 4.12 (Off-classical residual source). *Under Theorem 4.4 and Assumption 4.11,*

$$\Delta_{\text{cl}}\mathcal{S} = 0. \quad (64)$$

Hence $\mathcal{S} = \nu(\mathcal{E}(\rho_0) - \rho_0)$ is purely off-diagonal in the Everett pointer basis: an additive residual drive of off-classical structure, not a diagonal population source.

Proof. From (57) and (62),

$$\Delta_{\text{cl}}\mathcal{S} = \nu\Delta_{\text{cl}}(\mathcal{E}(\rho_0) - \rho_0) = \nu(\Delta_{\text{cl}}\mathcal{E}(\rho_0) - \Delta_{\text{cl}}\rho_0) = 0,$$

using linearity of Δ_{cl} . □

Remark 4.13 (What the toy justifies). The toy justifies the split used in (56): decoherence / outgoing loss sits in $\mathcal{L}_D \subset \mathcal{L}$, while unentangled incoming quantum information that cannot move the large- n diagonal feeds $\mathcal{S}$ in the off-classical sector. The Hadamard / shared- ϕ picture above is the interpretational content of that split: outgoing bits distinguish branches; incoming bits, when unentangled and non-preferential, refill inter-branch entanglement. Together with Sec. 4.2, the steady leftover then scales as $J_{\text{in}}/J_{\text{out}}$. It does not claim that every microscopic environment is exactly Poisson with product ancillas, nor that (62) holds beyond the stated large- n modelling regime. Concrete normalizations of $\mathcal{S}_H$ in continuum gravity remain system-dependent.

4.4 Exact open-system response on a general Hilbert space

Let $\mathcal{L}$ generate a strongly continuous quantum dynamical semigroup on the trace class. Let ρ_0 be a stationary reference state, $\mathcal{L}\rho_0 = 0$, and write $\rho = \rho_0 + \delta\rho$.

Assumption 4.14 (Stable trace-zero response). On an invariant trace-zero subspace $\mathcal{X} \subset \mathcal{T}_1(\mathcal{H})$ assume

$$\|e^{t\mathcal{L}}X\|_1 \leq Me^{-\gamma t}\|X\|_1, \quad X \in \mathcal{X}, \quad M \geq 1, \quad \gamma > 0, \quad (65)$$

and let the source $\mathcal{S} \in \mathcal{X}$. Under the unentangled-incoming construction of Theorem 4.4, the driven perturbation obeys $\dot{\delta\rho} = \mathcal{L}\delta\rho + \mathcal{S}$ with $\mathcal{L}$ and $\mathcal{S}$ as in (57); more generally the same driven equation may be taken as the linear-response modelling hypothesis on $\mathcal{X}$.

Theorem 4.15 (Resolvent response). *On $\mathcal{X}$,*

$$-\mathcal{L}^{-1}X = \int_0^\infty e^{t\mathcal{L}}X \, dt, \quad \delta\rho_{\text{ss}} = -\mathcal{L}^{-1}\mathcal{S}. \quad (66)$$

Hence the source-induced force response is

$$F_{\text{resp}} = -\text{Tr}[F\mathcal{L}^{-1}\mathcal{S}], \quad (67)$$

and, for any separately specified diagnostic dephasing Δ ,

$$F_{\text{coh}}^{(\Delta),\text{ss}} = -\text{Tr}[F(I - \Delta)\mathcal{L}^{-1}\mathcal{S}]. \quad (68)$$

Proof sketch. Exponential stability gives Bochner convergence of the integral. The force identities follow by linearity of the trace pairing. $\square$

Theorem 4.16 (Nonzero drive yields a unique stable nonzero steady leftover). *Under Assumption 4.14, consider the driven linear equation*

$$\dot{\delta\rho} = \mathcal{L}\delta\rho + \mathcal{S} \quad (\mathcal{S} \in \mathcal{X}). \quad (69)$$

Then:

1. $\delta\rho_{\text{ss}} := -\mathcal{L}^{-1}\mathcal{S}$ is the unique steady state in $\mathcal{X}$ (i.e. the unique solution of $\mathcal{L}\delta\rho + \mathcal{S} = 0$).
2. Writing $\delta\rho(t) = \delta\rho_{\text{ss}} + \varepsilon(t)$, one has $\dot{\varepsilon} = \mathcal{L}\varepsilon$, so

$$\|\varepsilon(t)\|_1 \leq Me^{-\gamma t} \|\varepsilon(0)\|_1. \quad (70)$$

Hence $\delta\rho_{\text{ss}}$ is globally exponentially asymptotically stable on $\mathcal{X}$.

3. $\delta\rho_{\text{ss}} = 0$ if and only if $\mathcal{S} = 0$. In particular, a nonzero residual drive $\mathcal{S} \neq 0$ produces a unique nonzero stable steady leftover $\delta\rho_{\text{ss}} \neq 0$ (and therefore a candidate Hysterical force $F_H = -\text{Tr}(F\mathcal{L}^{-1}\mathcal{S})$ that vanishes only if the force pairing with that leftover vanishes).

Proof. (1) Existence: $\mathcal{L}\delta\rho_{\text{ss}} + \mathcal{S} = -\mathcal{S} + \mathcal{S} = 0$ by construction of $\mathcal{L}^{-1}$ on the stable subspace. Uniqueness: if $\mathcal{L}\delta_1 + \mathcal{S} = \mathcal{L}\delta_2 + \mathcal{S} = 0$, then $\mathcal{L}(\delta_1 - \delta_2) = 0$. Exponential stability forces $\ker \mathcal{L} = \{0\}$ on $\mathcal{X}$ (a nonzero kernel vector would be a nondecaying orbit), so $\delta_1 = \delta_2$.

(2) Substitute $\delta\rho = \delta\rho_{\text{ss}} + \varepsilon$ into the driven equation and use (1): $\dot{\varepsilon} = \mathcal{L}\varepsilon$. The bound is Assumption 4.14.

(3) If $\mathcal{S} = 0$ then $\delta\rho_{\text{ss}} = 0$. Conversely, $\delta\rho_{\text{ss}} = 0$ means $\mathcal{L}^{-1}\mathcal{S} = 0$; applying $\mathcal{L}$ (or using injectivity of $\mathcal{L}^{-1}$ on $\mathcal{X}$) yields $\mathcal{S} = 0$. $\square$

Remark 4.17 (Drive versus state). The residual source $\mathcal{S}$ is the *drive*, not the steady state. The stable steady leftover is $\delta\rho_{\text{ss}} = -\mathcal{L}^{-1}\mathcal{S}$; nonzero $\mathcal{S}$ implies nonzero $\delta\rho_{\text{ss}}$ under the standing invertibility on $\mathcal{X}$.

Remark 4.18 (Literature hypothesis: C_0 resolvent–Laplace identity). On a Banach space, if $T(t)$ is a C_0 -semigroup with generator A and growth bound ω_0 , then

$$R(\lambda, A) = \int_0^\infty e^{-\lambda t} T(t) dt \quad (\text{Re } \lambda > \omega_0) \quad (71)$$

[1, 2]. Under the exponential stability assumed here ($\omega_0 < 0$), this specializes to $(-A)^{-1} = \int_0^\infty T(t) dt$. We use that identity as a *literature hypothesis*: it is a standard theorem of semigroup theory, cited rather than re-proved. In the companion Lean development the stable Laplace integral is constructed and bounded directly; identifying it with the resolvent of an unbounded generator remains this literature hypothesis (not a silent gap). The formula is stated with the growth bound ω_0 , not the spectral bound $s(A)$; exponential stability of the semigroup is exactly the hypothesis we assume.

Remark 4.19 (Physical stationary state and unbounded forces). The total candidate stationary state $\rho_{\text{ss}} = \rho_0 - \mathcal{L}^{-1}\mathcal{S}$ must be positive as well as trace one. Positivity is automatic only when the source is incorporated through a completely positive physical driving model; for an abstract additive source it is an additional admissibility requirement. For unbounded F , the theorem is asserted only on an invariant domain where $X \mapsto \text{Tr}(FX)$ and the resolvent integral are well defined and finite.

Proposition 4.20 (Dynamical relevance). *An initial coherent perturbation X is force-invisible for all subsequent times precisely when $\text{Tr}[FP_{\text{coh}}e^{t\mathcal{L}}X] = 0$ for all $t \geq 0$. In finite dimension, if $\text{Tr}(FP_{\text{coh}}\mathcal{L}^k X) \neq 0$ for some k , then X is dynamically force-relevant.*

4.5 Source-channel susceptibility

Let (Ξ, μ) be a channel space and

$$\mathcal{S} = \int_{\Xi} J(\xi) S_{\xi} d\mu(\xi), \quad J(\xi) \geq 0. \quad (72)$$

If the integral may be interchanged with the admissible response functional, then

$$F_{\text{resp}} = \int_{\Xi} J(\xi) \chi(\xi) d\mu(\xi), \quad \chi(\xi) = -\text{Tr}[F\mathcal{L}^{-1}S_{\xi}]. \quad (73)$$

A coherence-resolved susceptibility is obtained, when desired, by replacing F with $X \mapsto \text{Tr}[F(I - \Delta)X]$. The generator may retain system-dependent dissipative parameters explicitly:

$$\mathcal{L} = \mathcal{L}_0 + \int \Gamma_{\text{fr}}(\eta) D_{\text{fr},\eta} d\nu(\eta) + \int \Gamma_{\text{em}}(\zeta) D_{\text{em},\zeta} d\lambda(\zeta). \quad (74)$$

Theorem 4.21 (Invariant channel sectors). *If $\mathcal{X} = \bigoplus_a \mathcal{X}_a$, $\mathcal{L}\mathcal{X}_a \subseteq \mathcal{X}_a$, and $\mathcal{S} = \sum_a J_a S_a$ with $S_a \in \mathcal{X}_a$, then*

$$F_{\text{resp}} = \sum_a J_a \chi_a, \quad \chi_a = -\text{Tr}[F(\mathcal{L}|_{\mathcal{X}_a})^{-1}S_a]. \quad (75)$$

Corollary 4.22 (Weak channel mixing). *If $\mathcal{L} = \mathcal{L}_0 + V$, $\mathcal{L}_0$ is block diagonal, the inverses are taken on the stable response subspace, and $\|\mathcal{L}_0^{-1}V\| < 1$, then*

$$\mathcal{L}^{-1} = \sum_{n=0}^{\infty} (-\mathcal{L}_0^{-1}V)^n \mathcal{L}_0^{-1}, \quad (76)$$

with

$$\|\mathcal{L}^{-1} - \mathcal{L}_0^{-1}\| \leq \frac{\|\mathcal{L}_0^{-1}\|^2 \|V\|}{1 - \|\mathcal{L}_0^{-1}V\|}. \quad (77)$$

Remark 4.23 (Lean status). Lean proves the operator Neumann series and remainder bound on a Banach space of continuous linear endomorphisms (`cor_weak_channel_mixing`, `cor_weak_channel_mixing_norm_bound`) together with the scalar majorant identity used in the estimate.

4.6 Random-phase arrivals

For

$$|\psi_\phi\rangle = \frac{|n\rangle + e^{i\phi}|f\rangle}{\sqrt{2}}, \quad \phi \sim \text{Unif}[0, 2\pi), \quad (78)$$

with orthonormal $\{|n\rangle, |f\rangle\}$ spanning a two-dimensional subspace $\mathcal{H}_2 \subseteq \mathcal{H}$, phase averaging gives

$$\sigma_{\text{in}} = \frac{1}{2\pi} \int_0^{2\pi} |\psi_\phi\rangle \langle \psi_\phi| \, d\phi = \frac{P_2}{2}, \quad (79)$$

where P_2 is the projector onto $\mathcal{H}_2$ (equivalently $I/2$ when working entirely inside that qubit). Therefore direct linear phase-sensitive coherence vanishes in the ensemble mean on $\mathcal{H}_2$. A phase-independent dissipative population alignment can nevertheless be generated later. Mean force must consequently be computed from the linear density-matrix dynamics, not inferred from a nonlinear coherence norm. At incorporation we assume the incoming degree of freedom is unentangled with the system, $\rho_{N+1}(0^+) = \rho_N \otimes \sigma_{\text{in}}$; subsequent dynamics may entangle it.

4.7 Symmetry and vector cancellation

Theorem 4.24 (Symmetry restriction). *Let a group act by unitaries U_g and spatial rotations R_g . If $U_g \rho U_g^\dagger = \rho$ and $U_g^\dagger F U_g = R_g F$, then $R_g \text{Tr}(\rho F) = \text{Tr}(\rho F)$ for every g . Hence the expected force lies in the invariant vector subspace. For axial symmetry it lies along the symmetry axis; for spherical geometry about a distinguished observation point it is radial.*

4.8 Finite-dimensional motivating example

The general response theory does not require $\dim \mathcal{H} < \infty$. The following calculation illustrates one mechanism only.

4.8.1 Scalar coherence does not control force

In a D -dimensional localization basis, $F_{\text{coh}} = \sum_{i \neq j} \rho_{ij} F_{ji}$. Define $O(\rho) = \sum_{i \neq j} |\rho_{ij}|$. There is no universal positive lower bound on $|F_{\text{coh}}|$ in terms of O alone.

Proposition 4.25 (Coherence norm does not control force). *For $D \geq 2$ and any fixed Hermitian F , there exist density matrices with $O > 0$ but $F_{\text{coh}} = 0$.*

Proof. The real vector space of Hermitian zero-diagonal matrices has dimension $D(D-1)$. The real linear functional $Q \mapsto \text{Tr}(QF)$ has a nontrivial kernel. Choose nonzero Hermitian zero-diagonal Q in that kernel and normalize its entrywise off-diagonal ℓ^1 norm to one. For sufficiently small $\epsilon > 0$, $\rho_\epsilon = I/D + \epsilon Q$ is positive. Then $O(\rho_\epsilon) = \epsilon$ while $F_{\text{coh}} = \epsilon \text{Tr}(QF) = 0$. $\square$

5 Hysterical neutrality and gravitational survival

5.1 Response framework developed above

Let $\mathcal{H}$ be a separable Hilbert space and let $\mathcal{T}_1(\mathcal{H})$ denote trace-class operators. The linearized open-system response is

$$\dot{\delta\rho} = \mathcal{L}\delta\rho + \mathcal{S}. \quad (80)$$

On the stable response subspace, the stationary correction is

$$\delta\rho_H = -\mathcal{L}^{-1}\mathcal{S}, \quad F_H = -\text{Tr}(F\mathcal{L}^{-1}\mathcal{S}). \quad (81)$$

Here F is the externally measured force observable for the *underlying interaction*. Equation (81) introduces no new force carrier.

Assumption 5.1 (Exponential response stability). For X in the stable trace-class response subspace,

$$\|e^{t\mathcal{L}}X\|_1 \leq M e^{-\gamma t} \|X\|_1, \quad M \geq 1, \quad \gamma > 0. \quad (82)$$

Lemma 5.2 (Resolvent susceptibility bound). *Under Assumption 5.1, for bounded F and trace-class $\mathcal{S}$,*

$$|F_H| \leq \frac{M}{\gamma} \|F\|_\infty \|\mathcal{S}\|_1. \quad (83)$$

Proof. On the stable subspace,

$$-\mathcal{L}^{-1}\mathcal{S} = \int_0^\infty e^{t\mathcal{L}}\mathcal{S} dt. \quad (84)$$

Trace duality and Eq. (82) therefore give

$$|F_H| \leq \int_0^\infty |\text{Tr}(F e^{t\mathcal{L}}\mathcal{S})| dt \quad (85)$$

$$\leq \|F\|_\infty \int_0^\infty \|e^{t\mathcal{L}}\mathcal{S}\|_1 dt \quad (86)$$

$$\leq \|F\|_\infty M \|\mathcal{S}\|_1 \int_0^\infty e^{-\gamma t} dt, \quad (87)$$

which is Eq. (83). $\square$

The factor M/γ is a response susceptibility scale. The theorem below fails if this susceptibility diverges rapidly enough to compensate neutralization. This is not a technical nuisance: it identifies a genuine critical-response loophole that must be tested in any application.

5.2 Neutrality must be defined for the composite channel

Classically we do not feel a long-range electromagnetic field from the Earth: it carries (to excellent approximation) as many protons as electrons, and opposite charges cancel. Electromagnetism is the familiar *motivating example* for the theorem below, not a special case carved into the mathematics. Macroscopically, in the present universe, one expects every non-gravitational long-range channel of ordinary bulk matter—electromagnetic, strong (confined colour), and weak (short-ranged)—to satisfy the external-force hypothesis of the theorem, each for its own physical reason. Gravity is different: every mass attracts, so the contributions add, and on large scales gravity dominates the composite body. The same inheritance applies to residual responses from quantum alternatives (informally, Everett branches): when a typical alternative looks force-quiet from outside, controlled Hysterical leftovers inherit that quietness.

A neutral macroscopic body is therefore not a collection of independently observed constituents. The relevant object is its externally visible composite interaction.

Definition 5.3 (Response-visible neutrality). Consider a sequence of composites indexed by size N . Let F_N^{ext} denote the force observable coupling the entire composite to a specified external probe, after internal degrees of freedom have been included in the channel description. The indicated operator norm is that of this probe-channel observable on the retained response space for the asymptotic question under study (for electromagnetism: typically the long-range / monopole sector seen by a distant test charge, not every near-zone multipole). The interaction is *response-visibly neutral* if

$$\|F_N^{\text{ext}}\|_{\infty} \longrightarrow 0 \quad (88)$$

in the macroscopic/asymptotic regime being studied.

This is intentionally stronger and cleaner than saying that a signed scalar charge happens to sum to zero. It is the externally measurable interaction that must neutralize. Charge neutrality is only a motivating reason why the electromagnetic instance of $\|F_N^{\text{ext}}\|_{\infty} \rightarrow 0$ may hold at long range; the theorem itself is stated for any interaction channel that meets (88) under controlled response. A surviving monopole field would mean the composite is *not* response-visibly neutral in that channel. Net charge zero alone is not a substitute for (88): multipoles or other residual couplings may keep $\|F_N^{\text{ext}}\|_{\infty}$ from vanishing unless the asymptotic probe and observable are chosen so that those contributions drop out.

Theorem 5.4 (Hysterical Neutrality Theorem). *For a sequence of composite channels, suppose:*

- (i) $\|F_N^{\text{ext}}\|_{\infty} \rightarrow 0$;
- (ii) $\|\mathcal{S}_N\|_1 \leq S_*$ uniformly;
- (iii) the stable semigroups satisfy Eq. (82) with $M_N/\gamma_N \leq C_*$ uniformly.

Then the corresponding Hysterical force satisfies

$$\boxed{F_{H,N}^{\text{ext}} \longrightarrow 0.} \quad (89)$$

More quantitatively,

$$|F_{H,N}^{\text{ext}}| \leq C_* S_* \|F_N^{\text{ext}}\|_{\infty}. \quad (90)$$

Proof. Apply Lemma 5.2 to each composite channel and use assumptions (ii)–(iii):

$$|F_{H,N}^{\text{ext}}| \leq \frac{M_N}{\gamma_N} \|F_N^{\text{ext}}\|_{\infty} \|\mathcal{S}_N\|_1 \leq C_* S_* \|F_N^{\text{ext}}\|_{\infty}. \quad (91)$$

Assumption (i) then gives the result. □

Remark 5.5 (What the theorem does not assume). The theorem does not assume a finite number of alternatives, bounded branch weights, a branch-counting measure, or a fundamental Hysterical field. It is a continuity theorem for an open-system response functional.

Remark 5.6 (Critical loophole). Neutrality need not be inherited if M_N/γ_N or $\|\mathcal{S}_N\|_1$ diverges quickly enough that their product overwhelms $\|F_N^{\text{ext}}\|_{\infty}$. Thus “ordinary neutrality implies Hysterical neutrality” is a theorem only in a controlled response phase. A divergent susceptibility is precisely the kind of behavior expected near a critical point and must not be silently excluded when studying the onset of macroscopic response.

5.3 Application remark: two-level force-contrast bound

In a two-level renewal reduction with force contrast δf between decoherence-selected alternatives, the steady leftover is controlled by a population bias B obeying

$$\boxed{|B| \leq \frac{|\delta f|}{2}}. \quad (92)$$

Thus $\delta f \rightarrow 0$ forces $B \rightarrow 0$ (and, under nondegenerate renewal weighting, forces the steady leftover to vanish) with no further restriction on splitting or temperature. We record only this compact formal statement; the pedagogical two-level gravitational toy is not developed here. The same bound shows why the order of coarse-graining matters: independently summing opposite-sign constituent biases need not cancel, whereas forming the composite force contrast first does.

5.4 Relation to symmetry cancellation

Symmetry remains a useful sufficient mechanism for establishing neutrality, but it is not the theorem itself.

Lemma 5.7 (Odd-observable cancellation). *If a unitary involution C leaves a stationary state invariant,*

$$C\rho_{\text{ss}}C^\dagger = \rho_{\text{ss}}, \quad (93)$$

and reverses an interaction force,

$$C^\dagger FC = -F, \quad (94)$$

then

$$\text{Tr}(\rho_{\text{ss}}F) = 0. \quad (95)$$

Proof. Cyclicity of the trace gives

$$\text{Tr}(\rho_{\text{ss}}F) = \text{Tr}(\rho_{\text{ss}}C^\dagger FC) = -\text{Tr}(\rho_{\text{ss}}F). \quad (96)$$

□

Thus a conjugation symmetry can establish exact cancellation for a neutral electromagnetic composite. The broader Hysterical Neutrality Theorem does not require that particular symmetry and therefore need not force the strong and weak interactions into the same microscopic explanation.

5.5 Interaction-specific physical inputs

The Neutrality Theorem is interaction-agnostic once $\|F_N^{\text{ext}}\|_\infty \rightarrow 0$ holds under controlled susceptibility. Electromagnetism motivates the statement; in macroscopic bulk matter today one expects the same external quietness—and therefore the same Hysterical inheritance—for every non-gravitational channel, each for its own reason.

5.5.1 Electromagnetism

Physical input. Ordinary bulk matter is often electrically neutral to high accuracy, and electromagnetic charge has both signs. That is the usual reason the electromagnetic instance of $\|F_N^{\text{ext}}\|_\infty \rightarrow 0$ holds at long range. Under controlled susceptibility, Theorem 5.4 then forces the Hysterical electromagnetic response to vanish with it.

5.5.2 Strong interaction

Physical input. QCD physical states are color singlets at macroscopic distances and color is confined, so the externally visible long-range color force vanishes. That vanishing is the neutrality input to which Theorem 5.4 applies. A dedicated gauge-theoretic treatment would be required to derive confinement itself from the response formalism.

5.5.3 Weak interaction

Physical input. The low-energy weak interaction is mediated by massive $W^\pm$ and Z bosons and is short-ranged. Its large-distance suppression is therefore not primarily an application of charge neutrality. A Hysterical correction that remains tied to the underlying weak propagating channel inherits that lack of an unscreened macroscopic range. The italicized condition is a modeling assumption, not proved here.

5.5.4 Gravity

Physical input. Ordinary nonrelativistic matter carries positive gravitational mass-energy and has no observed opposite-sign gravitational charge sector that neutralizes ordinary macroscopic bodies. Gravity is also long-ranged in the regime considered here. Therefore the neutralization hypothesis of Theorem 5.4 is not generically available to ordinary gravitating matter.

5.6 Gravitational survival as a corollary

Corollary 5.8 (Gravitational survival). *Gravity is not bound by Theorem 5.4 whenever the externally visible gravitational force fails to neutralize. The Neutrality Theorem therefore does not force $F_H \rightarrow 0$ in that channel, so a nonzero gravitational leftover is permitted. When a gravitational response channel is also nondegenerate (nonzero force contrast and energetic splitting, nondegenerate susceptibility), the force-contrast bound (92) is compatible with a nonzero population bias, and one may expect $F_H \neq 0$ if the measured pairing does not cancel for independent dynamical reasons.*

Remark 5.9. Corollary 5.8 is only escape from the neutrality hypothesis—permission, not a derivation of amplitude or macroscopic normalization. Under the positive-weight / monopole lower-bound hypotheses of the companion Lean specification, the same channel algebra already yields a strictly positive finite response witness; that is still a conditional lower bound, not a claim that Nature has entered the supercritical regime. Strict positivity of leftover force at a stable saturated fixed point is packaged later as Corollary 6.15, once the closed-loop order parameter sits on the supercritical pitchfork (or equivalent saturated end-state). Whether ordinary gravity ever crosses into that regime remains a physical input / deferred hypothesis, not proved here.

5.7 Ontology: response of an existing interaction

The fundamental mathematical structure is

$$\boxed{\text{underlying interaction} + \text{open-system dynamics} \longrightarrow \text{Hysterical modification of its response.}}$$

(97)

For an interaction X one may write schematically

$$F_X^{\text{obs}} = F_X + F_{H,X}, \quad (98)$$

with $F_{H,X}$ generated by $(F_X, \mathcal{L}_X, S_X)$. The same mechanism applies to every interaction in the framework; it does not proliferate independent microscopic “Hysteria fields.”

5.7.1 Why macroscopic field language remains legitimate

In applications one may introduce a smooth effective acceleration or potential,

$$\mathbf{g}_H = -\nabla\Phi_H, \quad (99)$$

and sometimes an effective density through

$$\nabla^2\Phi_H = 4\pi G\rho_H^{\text{eff}}. \quad (100)$$

These equations are a continuum closure for a coarse-grained gravitational response. Accordingly:

The term *Hysterical Response field* denotes an effective macroscopic representation of the Hysterical modification of the gravitational interaction.

6 Critical instability

The neutrality theorem of Section 5 assumes controlled susceptibility. The present section treats neutrality and instability as the subcritical and supercritical regimes of one closed-loop response problem.

6.1 Definitional interaction universality

Let X denote any interaction to which the response decomposition is applied, with interaction observable F_X , open-system generator $\mathcal{L}_X$, and admissible response source $\mathcal{S}_X$. Hysterical Response is *defined* to be the residual/open-system component of the response of this already-existing interaction. In the steady linear-response representation developed above,

$$\boxed{F_{H,X} := -\text{Tr}\left[F_X\mathcal{L}_X^{-1}\mathcal{S}_X\right]}. \quad (101)$$

Whenever exponential stability permits the resolvent representation,

$$F_{H,X} = \int_0^\infty \text{Tr}\left[F_X e^{t\mathcal{L}_X}\mathcal{S}_X\right] dt. \quad (102)$$

No new microscopic force, mediator, or interaction-specific “Hysteria field” is added by these equations. They isolate a component of the response of F_X itself.

Definition 6.1 (Interaction universality of Hysterical Response). For every interaction X in the domain of the response decomposition, the quantity $F_{H,X}$ defined by (101) is called the Hysterical component of the response of X . Whether $F_{H,X}$ is nonzero, long-lived, or macroscopic is decided by the Neutrality Theorem and the closed-loop dynamics of Section 6.

Remark 6.2 (What remains physical). Once an interaction is represented in the present framework, no additional “universality hypothesis” is required. The values and signs of susceptibilities, and the faithfulness of a chosen open-system model to a particular microscopic quantum field theory, remain physical inputs.

6.2 Local response structure near neutrality

Let $\mathbf{Y} \in \mathbb{R}^m$ denote a vector of externally visible charges, order parameters, or force-producing coordinates. Let $\mathbf{h} \in \mathbb{R}^m$ denote the corresponding effective Hysterical response coordinates after projection onto the same response sector.

Assumption 6.3 (Exact conjugation symmetry). There is a local involution under which $(\mathbf{Y}, \mathbf{h}) \mapsto (-\mathbf{Y}, -\mathbf{h})$. Consequently $\mathbf{Y} = \mathbf{h} = 0$ is an exact neutral state and the local constitutive maps are odd.

Thus, near neutrality,

$$\mathbf{h} = \chi_H \mathbf{Y} + O(\|\mathbf{Y}\|^3), \quad (103)$$

$$\mathbf{Y}_{\text{eq}}(\mathbf{h}) = \chi_R \mathbf{h} + O(\|\mathbf{h}\|^3). \quad (104)$$

The matrices χ_H and χ_R are respectively the Hysterical and ordinary interaction susceptibilities.

Assumption 6.4 (Relaxational closure). The ordinary interaction relaxes toward $\mathbf{Y}_{\text{eq}}(\mathbf{h})$ according to

$$\dot{\mathbf{Y}} = -\Gamma[\mathbf{Y} - \mathbf{Y}_{\text{eq}}(\mathbf{h})] + O(\|\mathbf{Y}\|^3), \quad (105)$$

where Γ is positive stable; in the reciprocal class used for the simple eigenvalue criterion below it is symmetric positive definite.

Definition 6.5 (Closed-loop Hysterical gain). The dimensionless linear gain is

$$\boxed{\mathbf{K} := \chi_R \chi_H}. \quad (106)$$

Substitution of (103)–(104) into (105) gives

$$\dot{\mathbf{Y}} = J\mathbf{Y} + O(\|\mathbf{Y}\|^3), \quad \boxed{J = -\Gamma(I - \mathbf{K})}. \quad (107)$$

6.3 The master neutrality–instability theorem

Theorem 6.6 (Hysterical Neutrality–Instability Theorem). *Under Assumptions 6.3–6.4, the neutral state $\mathbf{Y} = \mathbf{h} = 0$ is an exact fixed point. Let*

$$\alpha(J) := \max\{\text{Re } \lambda : \lambda \in \text{spec}(J)\} \quad (108)$$

be the spectral abscissa of the linearized closed-loop generator $J = -\Gamma(I - \mathbf{K})$. Then:

- (i) *if $\alpha(J) < 0$, neutrality is locally linearly asymptotically stable;*
- (ii) *if $\alpha(J) > 0$, neutrality is linearly unstable and there exists an arbitrarily small perturbation that grows exponentially at linear order;*
- (iii) *if $\alpha(J) = 0$, the system lies on the linear critical surface and linear theory alone does not decide nonlinear stability.*

Proof. Exact neutrality follows from oddness of the constitutive maps. Equation (107) is the Fréchet linearization at the fixed point. In finite dimension every solution of the linearization $\dot{\mathbf{y}} = J\mathbf{y}$ is a linear combination of Jordan modes $t^k e^{\lambda t} \mathbf{v}$ associated with eigenvalues $\lambda \in \text{spec}(J)$. Writing $\alpha(J) = \max \text{Re } \lambda$, one therefore has the elementary alternatives:

- (i) *if $\alpha(J) < 0$, every mode decays exponentially, so the origin is locally linearly asymptotically stable;*

- (ii) if $\alpha(J) > 0$, some mode grows as $e^{\alpha t}$ (up to a polynomial prefactor), so an arbitrarily small projection onto that generalized eigenspace is linearly unstable;
- (iii) if $\alpha(J) = 0$, center or purely oscillatory modes remain and linear theory alone does not decide nonlinear stability.

(The same trichotomy is the finite-dimensional linear-stability classification in standard ODE texts [12].) $\square$

Definition 6.7 (Three response phases). We call $\alpha(J) < 0$ the *inert Hysterical-neutral phase*, $\alpha(J) = 0$ the *Hysterical critical surface*, and $\alpha(J) > 0$ the *Hysterical-unstable phase*.

Corollary 6.8 (Reciprocal gain criterion). *Suppose there exists a positive-definite susceptibility metric in which Γ is positive definite and $\mathbf{K}$ is self-adjoint, with the linearized dynamics similar to*

$$-\Gamma^{1/2}(I - \widetilde{K})\Gamma^{1/2}, \quad (109)$$

where $\widetilde{K}$ is self-adjoint and has the same relevant gain eigenvalues. Then

$$\lambda_{\max}(\widetilde{K}) < 1 \implies \text{inert neutrality}, \quad (110)$$

$$\lambda_{\max}(\widetilde{K}) = 1 \implies \text{criticality}, \quad (111)$$

$$\lambda_{\max}(\widetilde{K}) > 1 \implies \text{linear instability}. \quad (112)$$

In the commuting symmetric case $\widetilde{K} = \mathbf{K}$.

Proof. The congruence by $\Gamma^{1/2}$ preserves the inertia of the self-adjoint factor. Hence the sign of the least restoring eigenvalue changes precisely when the largest gain eigenvalue crosses unity. $\square$

Remark 6.9 (Why the general theorem is stated using $\alpha(J)$). For non-normal, nonreciprocal response matrices, $\rho(\mathbf{K})$ or $\lambda_{\max}(\mathbf{K})$ alone need not characterize stability because Γ and $\mathbf{K}$ need not commute. The spectral abscissa of the actual Jacobian is the exact linear criterion. The unity-gain criterion is a corollary for the reciprocal/detailed-balance class.

6.4 Connection to the Liouvillian

The constructive theory gives the stationary response

$$\delta\rho_H = -\mathcal{L}^{-1}\mathcal{S}. \quad (113)$$

If the source is odd in the order parameter,

$$\mathcal{S}(\mathbf{Y}) = \sum_a Y_a \mathcal{S}_a + O(\|\mathbf{Y}\|^3), \quad (114)$$

and O_b are projected response observables, then

$$(\chi_H)_{ba} = -\text{Tr}\left(O_b \mathcal{L}^{-1} \mathcal{S}_a\right). \quad (115)$$

Thus the master theorem is directly a theorem about the feedback generated by the response susceptibility once a physical interaction supplies χ_R and Γ .

6.4.1 Controlled response and neutrality

If the stable response semigroup satisfies

$$\|e^{t\mathcal{L}}X\|_1 \leq M e^{-\gamma t} \|X\|_1, \quad (116)$$

then the resolvent theory gives

$$\|\chi_H\| \lesssim \frac{M}{\gamma} C_{OS}, \quad (117)$$

where C_{OS} collects the relevant observable and source norms. Consequently, sufficiently strong relaxation bounds the loop gain.

Corollary 6.10 (Subcritical sufficient condition). *In any operator norm compatible with the chosen finite-dimensional projection, if*

$$\|\chi_R\| \frac{M}{\gamma} C_{OS} < 1, \quad (118)$$

then $\|\mathbf{K}\| < 1$. In the reciprocal class this is sufficient for inert neutrality.

The same susceptibility scale M/γ that enters the Neutrality Theorem therefore also bounds the closed-loop gain: controlled relaxation keeps the response subcritical, while a closing gap can push it through criticality.

6.4.2 Slow modes and the critical loophole

Let the relevant Liouvillian spectral decomposition be formally

$$\mathcal{L}R_j = -\gamma_j R_j, \quad (119)$$

with dual modes L_j . A scalar projected susceptibility has the form

$$\chi_H = \sum_j \frac{(O_H|R_j)(L_j|\mathcal{S}_Y)}{\gamma_j}. \quad (120)$$

For one dominant slow mode,

$$\boxed{\chi_H = \frac{\mathcal{R}_0}{\gamma_0} + O(1)}, \quad \mathcal{R}_0 = (O_H|R_0)(L_0|\mathcal{S}_Y). \quad (121)$$

If $\mathcal{R}_0 > 0$, then $\gamma_0 \rightarrow 0^+$ produces a positive divergent susceptibility. If the ordinary scalar susceptibility is $\chi_R > 0$, the leading critical gap is

$$\boxed{\gamma_c \simeq \chi_R \mathcal{R}_0}. \quad (122)$$

Thus the critical-response loophole retained in the Neutrality Theorem—divergent M/γ defeating neutralization—is the same slow-mode mechanism that drives the closed-loop gain through unity.

6.5 Scalar dynamical theorem

A useful realization keeps the Hysterical coordinate dynamical rather than eliminating it quasistatically:

$$\dot{Y} = -\Gamma Y + \alpha h, \quad (123)$$

$$\dot{h} = \sigma Y - \gamma h, \quad (124)$$

with $\Gamma, \gamma > 0$. The characteristic polynomial is

$$P(\lambda) = (\lambda + \Gamma)(\lambda + \gamma) - \alpha\sigma. \quad (125)$$

Theorem 6.11 (Scalar Hysterical Instability Theorem). *For (123)–(124) with $\Gamma, \gamma > 0$:*

$$\alpha\sigma < \Gamma\gamma \implies (Y, h) = (0, 0) \text{ is asymptotically stable,} \quad (126)$$

$$\alpha\sigma = \Gamma\gamma \implies \lambda_+ = 0, \quad (127)$$

$$\alpha\sigma > \Gamma\gamma \implies (0, 0) \text{ is linearly unstable.} \quad (128)$$

The growing exponent, when the discriminant is nonnegative, is

$$\boxed{\lambda_+ = -\frac{\Gamma + \gamma}{2} + \frac{1}{2}\sqrt{(\Gamma - \gamma)^2 + 4\alpha\sigma}.} \quad (129)$$

In particular, under the supercritical hypothesis $\alpha\sigma > \Gamma\gamma$ one has $(\Gamma - \gamma)^2 + 4\alpha\sigma > (\Gamma + \gamma)^2$, so the square root is real and $\lambda_+ > 0$.

Proof. The characteristic roots are

$$\lambda_{\pm} = -\frac{\Gamma + \gamma}{2} \pm \frac{1}{2}\sqrt{(\Gamma - \gamma)^2 + 4\alpha\sigma} \quad (130)$$

whenever the discriminant is nonnegative. Stability follows from $\text{tr} = -\Gamma - \gamma < 0$ and $\det = \Gamma\gamma - \alpha\sigma$: the origin is asymptotically stable when $\det > 0$, critical when $\det = 0$, and unstable when $\det < 0$. When $\alpha\sigma > \Gamma\gamma$ the discriminant exceeds $(\Gamma + \gamma)^2$, so λ_+ is real and positive. $\square$

The quasistatic susceptibility is $\chi_H = \sigma/\gamma$ and the ordinary response susceptibility is $\chi_R = \alpha/\Gamma$. Therefore

$$\boxed{G = \chi_R \chi_H = \frac{\alpha\sigma}{\Gamma\gamma},} \quad (131)$$

so Theorem 6.11 is exactly the scalar unity-gain criterion.

Corollary 6.12 (Finite threshold time). *If $\lambda_+ > 0$ and the initial condition has nonzero projection A_+ onto the unstable mode, then for any finite linear threshold $X_* > |A_+|$ the unstable component reaches it after*

$$\boxed{t_* = \frac{1}{\lambda_+} \log\left(\frac{X_*}{|A_+|}\right).} \quad (132)$$

Hence every nonzero unstable seed reaches a finite threshold in finite time; the delay diverges only logarithmically as the seed tends to zero.

6.6 Noise removes the need for an imposed seed

Physical reaction networks are stochastic. Add zero-mean noise to the ordinary coordinate:

$$dY = (-\Gamma Y + \alpha h)dt + \sqrt{2D} dW_t, \quad (133)$$

$$dh = (\sigma Y - \gamma h)dt. \quad (134)$$

The symmetry of the drift and noise implies that symmetric initial data retain zero ensemble mean.

Proposition 6.13 (Stochastic excitation of the unstable mode). *Assume $D > 0$, $\sigma \neq 0$, and $\lambda_+ > 0$. Let l_+ be a left eigenvector of the drift matrix for λ_+ and set $Z = l_+^T(Y, h)^T$. The Y component of l_+ is nonzero, so the reaction noise projects onto Z . After normalization,*

$$dZ = \lambda_+ Z dt + \sqrt{2D_+} dW_t, \quad (135)$$

with $D_+ > 0$, and for $Z(0) = 0$,

$$\mathbb{E}[Z^2(t)] = \frac{D_+}{\lambda_+} (e^{2\lambda_+ t} - 1). \quad (136)$$

Meanwhile $\mathbb{E}[Y] = \mathbb{E}[h] = 0$.

Proof. If the Y component of l_+ vanished, the left-eigenvector equations with $\sigma \neq 0$ would force the remaining component to vanish, contradicting nonzero l_+ . Ito's formula gives $d\mathbb{E}[Z^2]/dt = 2\lambda_+ \mathbb{E}[Z^2] + 2D_+$, whose solution is (136). Odd symmetry preserves zero first moments. $\square$

Thus an exactly symmetric ensemble can undergo instability without an externally imposed sign: the mean remains zero while the variance grows exponentially.

6.7 Nonlinear saturation and spontaneous sign selection

The linear theorem establishes loss of stability but not the final state. A generic scalar odd normal form near a supercritical pitchfork is

$$\dot{Y} = rY - uY^3 + O(Y^5), \quad u > 0. \quad (137)$$

Proposition 6.14 (Supercritical pitchfork). *For the cubic truncation of (137), $r < 0$ has one stable fixed point $Y = 0$. For $r > 0$, $Y = 0$ is unstable and two stable fixed points appear:*

$$Y_{\pm} = \pm \sqrt{r/u}. \quad (138)$$

The associated potential is $V(Y) = -rY^2/2 + uY^4/4$. In particular the positive branch $Y_+ = \sqrt{r/u}$ is a linearly stable nonzero fixed point, and oddness supplies the negative twin $Y_- = -Y_+$.

Corollary 6.15 (Gravitational positivity at a stable saturated node). *Assume the closed-loop gravitational order parameter obeys the supercritical odd normal form of Proposition 6.14 with $r > 0$ and $u > 0$, so the saturated end-state contains the stable fixed points $Y_{\pm} = \pm \sqrt{r/u}$. **Modeling assumption.** On the response sector of interest the measured leftover force is odd in the order parameter and positively aligned on the positive branch: $F_H = \kappa Y + O(Y^3)$ with $\kappa > 0$ (equivalently,*

the positive-channel pairing of Corollary 5.8's Lean witness), so the linearization at Y_+ does not cancel the sign of Y_+ . Then at the stable positive node,

$$\boxed{F_H|_{Y=Y_+} > 0}, \quad (139)$$

while the twin node yields $F_H|_{Y=Y_-} < 0$. In particular $|F_H|$ is bounded away from zero at either saturated node.

Remark 6.16 (What is not claimed). Corollary 6.15 does *not* assert that ordinary gravity is supercritical, that a cosmological or galactic seed drives r through zero, or that $F_H(t) > 0$ for all t from $t = -\infty$. It packages only the end-state mathematics: once the system sits in the supercritical saturated regime and the positive branch (or positive-channel pairing) is selected, gravitational Hysterical leftover force is strictly positive at that stable fixed point. Crossing into the regime remains a physical hypothesis deferred beyond this paper.

With additive noise,

$$dY = (rY - uY^3)dt + \sqrt{2D} dW_t, \quad (140)$$

the zero-current stationary density is

$$\boxed{P_{\text{st}}(Y) = \frac{1}{Z} \exp\left(\frac{rY^2}{2D} - \frac{uY^4}{4D}\right)}. \quad (141)$$

For $r > 0$ it is exactly symmetric and bimodal. Hence spontaneous sign selection does not require an explicit symmetry-breaking term.

6.8 Finite-temperature reaction corollary

Consider a two-state conjugate reaction coordinate with effective bias energy gh . Detailed balance may be parametrized by

$$W_{-\rightarrow+} = \Gamma e^{\beta gh}, \quad W_{+\rightarrow-} = \Gamma e^{-\beta gh}. \quad (142)$$

For $Y = p_+ - p_-$,

$$\dot{Y} = 2\Gamma [\sinh(\beta gh) - Y \cosh(\beta gh)], \quad (143)$$

and therefore

$$\boxed{Y_{\text{eq}}(h) = \tanh(gh/T), \quad \chi_R = \frac{g}{T}}. \quad (144)$$

If $h = \chi_H Y$ quasistatically, the exact fixed-point equation is

$$\boxed{Y = \tanh(GY), \quad G = \frac{g\chi_H}{T}}. \quad (145)$$

For $G \leq 1$, $Y = 0$ is the only solution. For $G > 1$, two nonzero symmetry-related solutions appear. Near threshold,

$$\boxed{Y_{\pm}^2 = \frac{3(G-1)}{G^3} + O((G-1)^2)}. \quad (146)$$

This finite-temperature model supplies a concrete nonlinear realization of Theorem 6.6; it does not establish that any particular microscopic interaction attains $G > 1$.

6.9 All interactions are subject to the same phase criterion

Definition 6.1 supplies the interaction-universal Hysterical component; Theorem 6.6 supplies its local phase classification when a response sector closes a feedback loop. There is therefore no separate conjecture that electromagnetic, strong, weak, or gravitational interactions “possess” Hysterical Response. For every interaction in the response framework, the substantive question is instead which of the following occurs in the physical state under study:

$$\boxed{\text{Hysterical component} \longrightarrow \begin{cases} \text{neutral/inert,} & \alpha(J) < 0, \\ \text{critical,} & \alpha(J) = 0, \\ \text{unstable/symmetry breaking,} & \alpha(J) > 0. \end{cases}} \quad (147)$$

A vanishing or neutralized Hysterical component is therefore a dynamical outcome inside the universal decomposition: under controlled susceptibility it is exactly what Theorem 5.4 predicts when the outside force is quiet. Instability requires the closed-loop spectral condition of Theorem 6.6.

6.10 Neutralization, interaction dependence, and gravitational survival

Microscopic universality and macroscopic survival are linked by the same theorem: each interaction’s reason for looking quiet on a typical branch is the neutrality input that decides whether its leftover survives.

Interaction	Ordinary macroscopic mechanism	Hysterical consequence under controlled response
Electromagnetic	charge-neutral composites / conjugation symmetry	inherits neutrality
Strong	color singlets and confinement	externally visible color response neutralizes
Weak	short range and massive mediators; reaction-specific neutral sectors	no generic long-range macroscopic survival
Gravity	positive mass-energy; no known opposite-sign ordinary mass sector	neutrality hypothesis generally fails

Corollary 6.17 (Gravitational survival from failure of neutrality). *Gravity escapes Theorem 5.4 for ordinary positive-mass composites whenever the externally visible gravitational force fails to neutralize, so a nonzero Hysterical response is permitted and one may expect $F_H \neq 0$ when a concrete nondegenerate channel supplies a nonzero measured pairing. Once such a channel sits in the supercritical saturated regime of Proposition 6.14, Corollary 6.15 upgrades permission to strict positivity of F_H at the stable positive node under the stated positive-alignment hypothesis.*

The resulting architecture is therefore

$$\boxed{\text{Hysterical component of each interaction (definition)} \longrightarrow \begin{cases} \text{neutral/inert,} & \alpha(J) < 0, \\ \text{critical,} & \alpha(J) = 0, \\ \text{unstable/symmetry breaking,} & \alpha(J) > 0, \end{cases}} \quad (148)$$

followed, after the system settles into its macroscopic state, by interaction-specific neutralization. Gravity is exceptional not because it alone has microscopic Hysterical Response, but because ordinary positive mass-energy does not supply the neutralizing sign structure available to other interactions.

6.11 Density is a physical control parameter, not the theorem variable

The master theorem is spectral, not a density theorem. In a collision-dominated environment one often has schematically

$$\gamma_{\text{coll}} \sim n \langle \sigma v \rangle. \quad (149)$$

If this rate controls the relevant Hysterical mode and the residue varies slowly, then

$$n \downarrow \implies \gamma \downarrow \implies \chi_H \uparrow \implies G \uparrow. \quad (150)$$

Under those additional physical assumptions, diffuse states are naturally driven toward large Hysterical susceptibility while dense, rapidly equilibrating states tend toward inert neutrality.

A helpful visualization is a *soup* versus *meatballs* cartoon. Low-density, nearly homogeneous soup has a thin environment: residual open-system structure from quantum alternatives has more time to matter, so the medium behaves *more quantumly* in the limited sense that history-dependent leftovers can tip the closed-loop gain. Dense meatballs have a thick environment: fast classicalization and relaxation drown those leftovers, so locally the response looks *more classical*—ordinary dense matter near inert neutrality. The meatballs are better read as a saturated end-state after growth than as “the unstable phase.” The quantitative object behind the picture remains $G = \chi_R \chi_H$ (or $\alpha(J)$), not a quantum-versus-classical label on the equation of state.

Remark 6.18 (Critical slowing down at high density). Low density is not necessary. A dense plasma can possess a collective mode whose relaxation gap becomes small near a phase transition. The theorem follows the relevant eigenvalue of the response generator, not the total collision rate of the medium.

6.12 How to apply the instability theorems

To use the generic instability results in a concrete model it is enough to specify physical variables and identify the linearized Jacobian as an instance of Theorem 6.6 or Theorem 6.11.

For example, if a model has a matter perturbation δ and response coordinate q with

$$\dot{q} + \gamma q = \sigma \delta, \quad (151)$$

and the matter equation feeds q back into the growth of δ , one should: (i) construct the finite-dimensional first-order state vector; (ii) identify the resulting closed-loop Jacobian; and (iii) invoke Theorem 6.6 when its spectral abscissa is positive. Growth rates, thresholds, and observational consequences are then properties of that model. The generic facts that a positive eigenvalue gives exponential growth, finite threshold time for a nonzero projection, and stochastic excitation when noise projects onto the unstable mode are already supplied here.

6.13 What is proved and what remains physical

The mathematical results of this paper are:

1. exact neutrality and its complete local linear classification by $\alpha(J)$;
2. the reciprocal unity-gain criterion;
3. the relation of the gain to the Liouvillian susceptibility;
4. the one-slow-mode critical scaling $\chi_H \sim \mathcal{R}_0/\gamma_0$;
5. the scalar instability condition $\alpha\sigma > \Gamma\gamma$ and its exact growth exponent;
6. finite threshold time for nonzero unstable projection;
7. stochastic excitation of an unstable mode without a preferred sign;

8. generic pitchfork saturation with stable nonzero $\pm$ nodes (Lean); the finite-temperature $Y = \tanh(GY)$ fixed-point formula is an explicit physical realization, not a separately Lean-certified theorem;
9. gravitational positivity at the stable positive saturated node under positive-channel / positive-alignment hypotheses (Lean `cor_gravity_positivity`), without proving that gravity enters the supercritical regime.

Interaction universality is not on this list: within the response framework it is Definition 6.1. The following remain physical inputs or modeling questions rather than consequences of the theorem: whether a chosen open-system representation faithfully captures a particular microscopic quantum field theory; the values and signs of its susceptibilities; the dependence of the relevant Liouvillian spectrum on density, temperature, and cosmological epoch; whether a given system actually crosses the critical surface (gravitational seeding / criticality); the nonlinear macroscopic amplitude after a particular physical instability beyond the normal-form packaging; and the empirical claim that ordinary gravity lacks a macroscopic positive-mass neutralizing sector in the regime of interest.

7 Infrared equivalence

Weak-field macroscopic predictions depend on the retained Hysterical potential Φ_H , not on whether gravity is treated as semiclassical geometry sourced by quantum matter or as quantum before coarse graining. Both routes define the gravitational Hysterical component by the same response construction (81) applied to an existing gravitational observable; neither invents a second gravitational interaction.

Assumption 7.1 (Weak-field infrared limit). On scales L in the regime of interest, the retained gravitational response admits an effective metric

$$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + h_{\mu\nu}^{\text{ord}} + h_{\mu\nu}^H + r_{\mu\nu}^{\text{UV}}, \quad (152)$$

with $\|h^{\text{ord}}\|, \|h^H\| \ll 1$ and a remainder satisfying

$$\frac{\|r^{\text{UV}}\|}{\|h^{\text{ord}}\| + \|h^H\|} \rightarrow 0 \quad (153)$$

as the infrared limit is taken within the stated regime.

Assumption 7.2 (Newtonian scalar sector). For slowly moving probes and quasi-static sources, the leading retained Hysterical acceleration is conservative on the domain $\Omega \subseteq \mathbb{R}^3$:

$$\mathbf{g}_H = -\nabla\Phi_H, \quad (154)$$

with $\Phi_H \in C^2(\Omega)$.

Assumption 7.3 (Finite coarse-grained response). The microscopic response has a finite retained infrared projection, so that Φ_H exists at the order kept in Assumption 7.1.

Assumption 7.4 (Common ordinary infrared gravity). Both microscopic descriptions reproduce the same leading ordinary weak-field gravitational law on the domain being compared.

Definition 7.5 (Effective Hysterical density). For any C^2 potential Φ_H satisfying Assumption 7.2, define

$$\boxed{\rho_H^{\text{eff}} := \frac{1}{4\pi G} \nabla^2 \Phi_H.} \quad (155)$$

Lemma 7.6 (Effective-source representation). *Under Assumption 7.2,*

$$\boxed{\mathbf{g}_H = -\nabla\Phi_H, \quad \nabla^2\Phi_H = 4\pi G\rho_H^{\text{eff}}.} \quad (156)$$

Proof. The first identity is Assumption 7.2. The second is Definition 7.5. No claim that ρ_H^{eff} is a new microscopic material density is required. $\square$

Definition 7.7 (Infrared response equivalence). Descriptions A and B satisfy $A \sim_{\text{IR}} B$ to retained order if, after coarse graining onto the common weak-field domain,

$$\Phi_H^A - \Phi_H^B = r_H, \quad \|r_H\|_{C^2} = o(\epsilon), \quad (157)$$

where ϵ denotes the retained weak-field response order.

Theorem 7.8 (Hysterical Gravitational Infrared Equivalence Theorem). *Suppose descriptions A and B (for example a semiclassical theory and a quantum-gravitational theory) satisfy Assumptions 7.1–7.4 on the same weak-field domain and are infrared response equivalent in the sense of (157). Then, to retained order:*

- (i) $\mathbf{g}_H^A - \mathbf{g}_H^B = o(\epsilon)$;
- (ii) $\rho_{H,A}^{\text{eff}} - \rho_{H,B}^{\text{eff}} = o(\epsilon)$;
- (iii) *every leading weak-field observable that is a continuous functional of Φ_H and its derivatives through second order agrees to the same retained order.*

Consequently, once the retained potentials agree, the leading macroscopic Hysterical gravitational predictions are independent of whether the microscopic description is semiclassical or fundamentally quantum.

Proof. From (157) and $\mathbf{g}_H = -\nabla\Phi_H$, $\mathbf{g}_H^A - \mathbf{g}_H^B = -\nabla r_H = o(\epsilon)$. Applying Definition 7.5 gives $\rho_{H,A}^{\text{eff}} - \rho_{H,B}^{\text{eff}} = (4\pi G)^{-1}\nabla^2 r_H = o(\epsilon)$. The final statement follows by continuity of any retained observable functional in the C^2 topology. $\square$

Corollary 7.9 (Exact infrared agreement). *If $\Phi_H^A = \Phi_H^B$ on the domain, then $\mathbf{g}_H^A = \mathbf{g}_H^B$ and $\rho_{H,A}^{\text{eff}} = \rho_{H,B}^{\text{eff}}$ exactly there.*

Remark 7.10 (What the equivalence does not require). The two microscopic density operators, Hilbert spaces, generators, or ultraviolet field contents need not be equal. Only the retained infrared gravitational response must agree. This is an equivalence of phenomenology at a specified order, not an equivalence of microscopic theories.

7.1 Static dephasing and dynamical response are logically distinct

The static coherence contribution relative to a dephasing map Δ is

$$F_{\text{coh}}^{(\Delta)}(\rho) = \text{Tr}[F(I - \Delta)\rho]. \quad (158)$$

If $\Delta^\dagger(F) = F$, then $F_{\text{coh}}^{(\Delta)}(\rho) = 0$. This statement is sometimes tempting to overinterpret.

Proposition 7.11 (No-go independence). *The condition $\Delta^\dagger(F) = F$ and the vanishing of $F_{\text{coh}}^{(\Delta)}$ do not, by themselves, imply*

$$F_H = -\text{Tr}[F\mathcal{L}^{-1}\mathcal{S}] = 0. \quad (159)$$

Indeed, there exist finite-dimensional choices for which the static coherence term vanishes while $F_H \neq 0$.

Proof. Take a two-dimensional real diagonal sector represented by vectors. Let $F = (1, 0)$ and let Δ be the identity map. Then the static off-diagonal/coherence contribution vanishes identically. Let $\mathcal{L} = -I$ and $\mathcal{S} = (1, 0)$. Hence $\mathcal{L}^{-1} = -I$ and

$$F_H = -F \cdot \mathcal{L}^{-1} \mathcal{S} = F \cdot \mathcal{S} = 1 \neq 0. \quad (160)$$

Thus the static dephasing condition is not sufficient to annihilate the dynamical response functional. The example proves logical independence; physical models impose additional structure. $\square$

Corollary 7.12 (Semiclassical decoherence objection). *Elimination of a chosen static off-diagonal expectation value does not establish elimination of the integrated open-system response. A semiclassical calculation must evaluate the response dynamics, not infer $F_H = 0$ solely from dephasing.*

Remark 7.13 (Infrared inheritance). Any retained weak-field observable that is a continuous functional of Φ_H (accelerations, effective densities, local weak-field tests) is therefore shared by infrared-equivalent microscopic descriptions. The theorem does not apply if the descriptions lack a common weak-field limit, if the response lacks a finite infrared projection, if the leading response is intrinsically nonconservative beyond the scalar approximation, or if ultraviolet corrections remain of the same order as the claimed effect. Conversely, a vanishing gravitational Hysterical susceptibility, or observations that exclude the resulting Φ_H , constrain the physical realization: infrared agnosticism about quantization is not immunity from empirical testing.

8 Conclusion

The picture is simple to state, even if the proofs are not. Decoherence erases coherent interference among quantum alternatives from the reduced description; a history-dependent leftover can remain on ordinary forces. That leftover is covariant under passive renaming of the basis once the environment's classical map transforms with everything else. When a whole body is force-neutral from the outside and the response stays under control, the leftover vanishes with the outside force—so electromagnetism, colour, and the weak interaction can be hysterically quiet on large scales. Gravity keeps a nonzero outside force for ordinary positive-mass bodies, so the Neutrality Theorem leaves a gravitational leftover in play. If that leftover feeds back hard enough, the quiet neutral point can tip into growth instead of decay; in the supercritical saturated end-state the odd pitchfork supplies stable nonzero nodes, and under positive-channel alignment the positive node carries strictly positive leftover force. For the weak-field infrared story, semiclassical and quantum-gravitational accounts share the predictions once they keep the same leftover potential. Subsequent galactic modelling continues that gravitational Hysterical Response into a spatial halo; it does not introduce a new force carrier.

What remains for physics or further modelling includes how strong the gravitational leftover is, how the environment chooses its classical map, and whether real systems actually enter or sit near a supercritical tipping regime.

Lean status and literature hypotheses

Companion Lean specifications live under `HystericalResponse/Paper1/`, structured as the TeX section route (modules and theorem names follow the paper; the umbrella import file is the map). They are faithful ports of the TeX proofs at the stated algebraic / Banach abstraction; `lake build` of the companion umbrella module is the machine check. In particular the unentangled-incoming driven-deviation and off-classical-source identities, the $J_{\text{in}}/J_{\text{out}}$ resolvent scaling wrapper,

the unique stable steady leftover for nonzero drive (including the exponential envelope for deviations), constructive response lemmas including the weak-mixing operator Neumann series and remainder bound, neutrality inheritance, gravitational survival / positive-channel lower bounds, reciprocal/mode closed-loop criteria, scalar $\lambda_{\pm}$ signs, threshold hitting time, noise variance ODE, supercritical pitchfork saturation with stable $\pm$ nodes and gravitational positivity at the positive saturated node under positive alignment, and infrared Lipschitz/linear observable continuity are Lean-proved. The following are *not* claimed as Lean-proved originals:

- the unbounded C_0 generator resolvent–Laplace identification of Remark 4.18 [1, 2];
- collisional decoherence rates $\propto$ environmental flux [8, 9, 10] and the standard status of driven / inhomogeneous master equations [5, 11] (Remarks 4.5–4.8);
- the Jordan-mode linear ODE stability dictionary used when reading Theorem 6.6’s spectral abscissa $\alpha(J)$ as asymptotic stability of $\dot{y} = Jy$ for a general finite-dimensional generator [12] (the elementary α trichotomy and the reciprocal/scalar reductions are proved directly);
- physical inputs such as applicability of a chosen open-system model to Nature, the large- n diagonal-protection modelling hypothesis, the identification $\gamma \sim J_{\text{out}}$, and continuum normalizations of $\mathcal{S}_H$.

Remark 8.1 (Unrelated). The collective noun for Sugar Tongs is a “Monopsony.”

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